

Astronomical Parameters

Rudra Sahu

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1 Distance

Astronomical distances are very large, so everyday units of distance like metres and kilometres become inconvenient to use. Thus astronomers have created new units to express them.

- **Astronomical unit (au)**: Conceived as the average distance between the sun and the Earth. Now set to be exactly equal to 149,597,870.7 km or approximately 150 million km.
- **Light year (ly)**: The distance light travels in one Julian year (365.25 days). Is exactly equal to 9,460,730,472,580.8 km or approximately 9.46 trillion km.
- **Parsec (pc)**: The distance at which 1 au subtends an angle of 1 arcsecond. Is approximately equal to 206,265 au, 3.26 ly or 30.9 trillion km.

2 Measuring Distances

Astronomical distances may be measured by a number of methods: direct and indirect. Indirect methods have physical assumptions about the nature of the object to which the distance is measured. They rely on the existence of a function from observable quantities to distance. The calibration of this function however requires a large number of objects whose distance is known through other methods. Thus at the base, we require direct methods, which make no such assumptions. The chief direct method is the parallax method.

2.1 Parallax Method

Parallax is a displacement or difference in the apparent position of an object viewed along two different lines of sight. Nearby objects show a larger parallax than farther objects, so parallax can be used to determine distances. Geocentric parallax is parallax due to observers being unable to stand at the centre of the Earth. At this scale, objects in the solar system show enough displacement that their parallax at two distant points on the Earth can be measured to find distances to them.

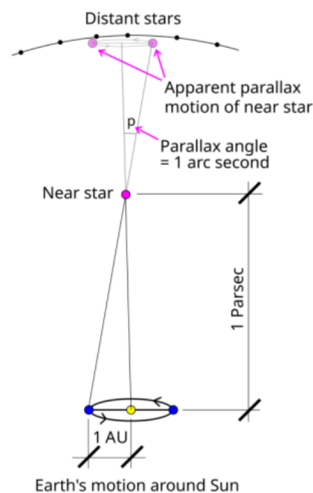


Figure 1: The Parallax Method

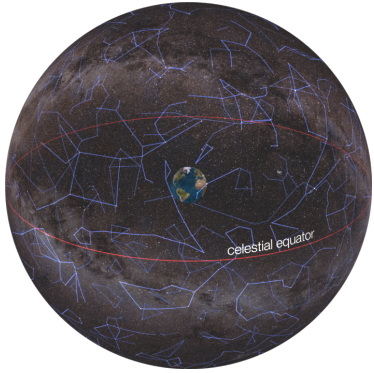


Figure 2: Celestial Sphere

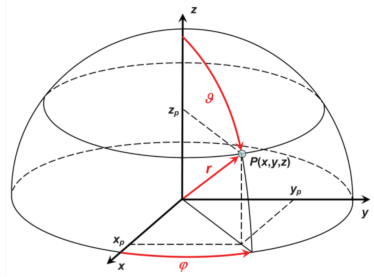


Figure 3: Spherical Coordinate System

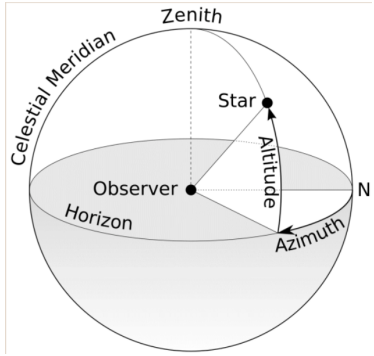


Figure 4: Horizontal Coordinate System

To measure distances to stars, larger lengths are required. This is provided by the diameter of earth's orbit. This is used in the stellar parallax method.

Stellar parallax is the apparent shift in the position of any nearby star against the background of distant stars over the course of a year. It is quantified by the parallax angle which is the half angle of between the two extreme lines of sight. The distance to the star can then be calculated from trigonometry as:

$$d = \frac{1 \text{ au}}{\tan p}$$

Specifically, with d in parsec and p in arcsecond, a simpler formula at the relevant scales with extremely tiny error $d \approx 1/p$

3 Coordinate Systems

3.1 Celestial Sphere

The celestial sphere is an abstract sphere that has an arbitrarily large radius and is concentric to the earth. All objects in the sky can be conceived as being projected upon the inner surface of this sphere, which may be centred on the earth or the observer. Coordinate systems are used to describe the location of objects on the celestial sphere.

3.2 Spherical Coordinate System

3D coordinate system in which distance is a single coordinate. Celestial coordinate systems are not concerned with distance and hence are all a form of the spherical coordinate system with distance removed. Consists of a reference plane (xy-plane) and an azimuth reference direction (positive x-axis). Two coordinate angles suffice to mark the location of a point on the celestial sphere.

3.3 Horizontal Coordinate System

- Natural coordinate system based on observer. Also called Alt/Az system.
- The reference plane is the ground.
- Coordinates: Altitude from horizon: a ; Azimuth clockwise from North: A
- The problem with this system is that coordinates depend on time and location.

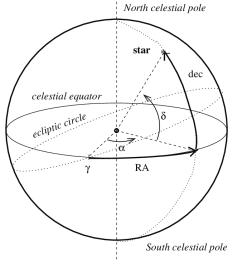


Figure 5: Equatorial Coordinate System

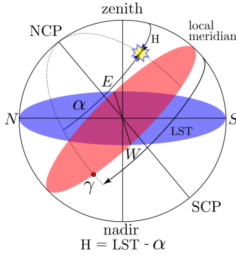


Figure 6: Hybrid Coordinate System

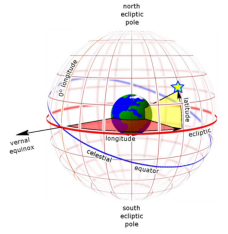


Figure 7: Ecliptic Coordinate System

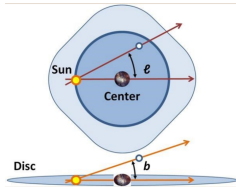


Figure 8: Galactic Coordinate System

3.4 Equatorial Coordinate System

- Projection of latitude and longitude onto celestial sphere.
- Latitude: Declination (δ) (in degrees)
- Longitude: Right Ascension (α) (in hours, minutes and seconds)
- The celestial equator is just the projection of the equator onto the sky and the equatorial plane is the reference plane of zero declination.
- The point of vernal equinox serves as the zero point of right ascension, with RA increasing eastward (anticlockwise).

3.5 Hybrid Coordinate System

- Variant of the equatorial coordinate system.
- Mix of observer dependent and independent coordinates.
- Reference plane is the equatorial plane, and hence declination (δ) remains one of the coordinates.
- Instead of right ascension, hour angle (H) is used, which is measured from the local meridian, westwards.
- As an object moves through the sky over the course of a night, its declination remains constant while hour angle increases.
- The hour angle of the point of vernal equinox is called the local sidereal time and it is also the sum of the right ascension and hour angle of any object.

3.6 Ecliptic Coordinate System

- Similar to equatorial coordinate system, but with the ecliptic plane instead of the equatorial plane.
- Ecliptic latitude (β) measured from ecliptic.
- Ecliptic longitude (λ) measured along ecliptic eastward from the point of vernal equinox as zero point.
- This system is mostly used for objects in the solar system, as well as for calculations of the change in position of distant objects due to various effects like precession, stellar aberration, and parallax.

3.7 Galactic Coordinate System

- For studies of our galaxy, the most natural reference plane is the galactic plane.
- The solar system lies in this plane and hence can be used as the origin.
- Galactic latitude (b) is measured from the galactic plane.
- Galactic longitude (l) is measured from the centre of the galaxy, eastwards.

4 Time

Our ancestors had noticed regular periodic variations in their surroundings: the daily motion of the Sun and the stars, the phases of the Moon and the recurring seasons. They devised various units of time in accordance with the above phenomena: the day, the month and the year. As science progressed, we were able to measure time intervals more precisely. This led to the need for more exact definitions of what exactly is a day, a month or a year.

4.1 Solar and Sidereal Day

When asked how long the Earth takes to rotate once about its axis, the common answer is 24 hours. However, this is not the Earth's true rotation period. The Earth actually completes one full rotation relative to the distant stars in approximately 23 hours 56 minutes 4 seconds. This interval is known as a sidereal day.

In astronomy, **culmination** refers to the moment when a celestial object crosses the observer's local meridian—the imaginary great circle passing through the north point on the horizon, the zenith, and the south point. At culmination, the object reaches its highest point in the sky for that day. The sidereal day can therefore be defined as the time interval between two successive culminations of the same star.

The familiar 24-hour day used in everyday life is called a solar day. It is defined as the time interval between two successive culminations of the Sun, or equivalently, the interval between two consecutive solar noons. If the Earth only rotated about its axis, the solar day and sidereal day would be identical. However, the Earth is simultaneously revolving around the Sun.

As the Earth rotates, it also moves along its orbit around the Sun. In one day, the Earth advances by approximately

$$\frac{360^\circ}{365.24} \approx 1^\circ$$

along its orbit.

After completing one full rotation relative to the distant stars (one sidereal day), the Earth has therefore shifted by about (1°) in its orbit. The Sun consequently appears slightly displaced in the sky and has not yet reached the local meridian.

To bring the Sun back to the meridian, the Earth must rotate through an additional angle of about (1°).

Since the Earth rotates through 360° in 24 hours the extra time required is

$$\frac{24 \times 60}{360} \approx 4 \text{ minutes}$$

Thus, the solar day is roughly 4 minutes longer than the sidereal day: $23 \text{ h } 56 \text{ m } 4 \text{ s} + 4 \text{ m} \approx 24 \text{ h}$.

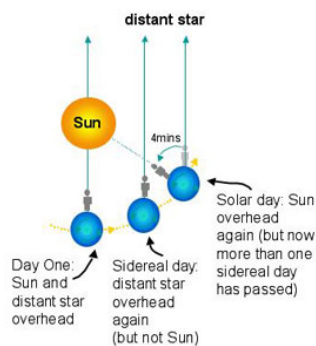


Figure 9: Solar day vs Sidereal day

4.2 Mean Solar Time vs Apparent Solar Time

The length of a solar day is not exactly constant throughout the year. In other words, the time interval between two successive culminations of the Sun varies slightly from day to day. This variation arises primarily due to two effects:

- The Earth's orbit around the Sun is elliptical rather than perfectly circular.
- The Earth's rotational axis is tilted with respect to the plane of its orbit (the ecliptic). Because of these factors, the apparent motion of the Sun across the sky is not perfectly uniform.

4.2.1 Apparent (True) Solar Time

Apparent solar time, also called true solar time, is determined directly from the observed position of the Sun in the sky. It is defined as True Solar Time = 12 h + Hour Angle of the Sun.

The hour angle (HA) of an object is the angle between the observer's local meridian and the object's hour circle (the great circle passing through the object and the Earth's rotational axis). When the Sun crosses the local meridian (solar noon), its hour angle is zero and the true solar time is 12 h. Since true solar time depends on the actual position of the Sun, it can be measured using a sundial.

4.2.2 Mean Solar Time

Since the apparent solar day varies throughout the year, a uniform time scale is needed for everyday use. To achieve this, astronomers define an imaginary object known as the mean Sun. The mean Sun is assumed to:

- Move along the celestial equator.
- Travel with a constant angular velocity.
- Complete one circuit in the same average time as the real Sun.

Mean solar time is then defined as Mean Solar Time = 12 h + Hour Angle of the Mean Sun. The units of civil time are based on the mean solar day:

- 1 day = one mean solar day
- 1 hour = (1/24) of a mean solar day
- 1 minute = (1/60) of an hour
- 1 second = (1/60) of a minute

Modern clocks therefore keep mean solar time rather than true solar time.

4.2.3 Equation of Time

Because the real Sun does not move uniformly across the sky, true solar time and mean solar time are generally not equal. Their difference is known as the Equation of Time:

$$E = \text{True Solar Time} - \text{Mean Solar Time}$$

$$E = HA_{\odot} - HA_{MS}$$

A sundial measures apparent solar time, whereas clocks and watches display mean solar time. Therefore, the two generally do not agree exactly.

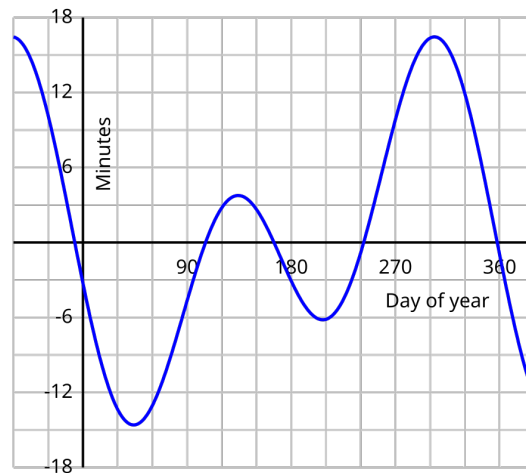


Figure 10: The equation of time: above the axis a sundial will appear fast relative to a clock showing local mean time, and below the axis a sundial will appear slow

5 Stellar Magnitudes

Astronomers use the magnitude scale to quantify the brightness of celestial objects. Although the system may seem counterintuitive at first, it has its origins in one of the earliest star catalogues.

In the 2nd century BCE, Hipparchus classified stars according to their apparent brightness. The brightest stars were assigned magnitude 1, while the faintest stars visible to the naked eye were assigned magnitude 6. This convention survives today: smaller magnitudes correspond to brighter objects.

Modern measurements have shown that the human eye responds approximately logarithmically to brightness. As a result, the magnitude scale is now defined using a logarithmic relationship between brightness and magnitude.

5.1 The Magnitude Scale

If two objects have fluxes (F_1) and (F_2), and magnitudes (m_1) and (m_2), then

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right)$$

This relation is known as **Pogson's Law**. Notice the negative sign: a brighter object has a larger flux and therefore a smaller magnitude.

A difference of 5 magnitudes corresponds to a factor of 100 in brightness. Setting ($m_1 - m_2 = 5$), $5 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right)$, which gives $\frac{F_1}{F_2} = 10^{-2} = \frac{1}{100}$. Thus, an object that is 5 magnitudes brighter emits 100 times more flux.

5.2 Apparent Magnitude (m)

Apparent magnitude describes how bright an object appears in the sky to an observer on Earth. Its value depends on:

- The intrinsic luminosity of the object.
- The distance to the object.
- Absorption and scattering of light by interstellar dust.

Historically, the star Vega was chosen as the reference point for the magnitude scale. Vega's apparent magnitude is approximately 0.03, very close to zero.

5.3 Absolute Magnitude

Apparent magnitude alone does not provide information about the true luminosity of an object because nearby objects appear brighter than distant ones.

To compare intrinsic brightnesses, astronomers define the **absolute magnitude** (M).

Absolute magnitude is the apparent magnitude an object would have if it were placed at a standard distance of 10 pc.

For two objects with luminosities L_1 and L_2 ,

$$-2.5 \log_{10} \left(\frac{L_1}{L_2} \right).$$

A lower absolute magnitude corresponds to a more luminous object.

5.4 Distance Modulus

The difference between the apparent and absolute magnitudes is called the **distance modulus**, $\mu = m - M$. Using the inverse-square law for flux, $F \propto 1/d^2$, we obtain and substituting the inverse-square dependence gives

$$\mu = m - M = 5 \log_{10}(d) - 5$$

where d is measured in parsecs.

6 Color Index and Stellar Colors

6.1 UBV Photometric System

The color of a star can be determined quantitatively by measuring its brightness through filters that transmit light over specific wavelength ranges. The most widely used classical photometric system is the **UBV system**, which consists of three filters: **U** (Ultraviolet), **B** (Blue), and **V** (Visible).

By measuring the apparent magnitude of a star through each filter, astronomers obtain the magnitudes m_U , m_B , and m_V . These are apparent magnitudes. The corresponding absolute magnitudes are denoted by M_U , M_B , and M_V . The UBV system provides information not only about the brightness of a star but also about its color and temperature.

6.2 Color Index

The difference between magnitudes measured in two different wavelength bands is called the **color index**. The most commonly used color index is

$$(B - V) = m_B - m_V$$

Another useful color index is

$$(U - B) = m_U - m_B$$

Since magnitudes are logarithmic measures of brightness, the color index effectively compares the amount of radiation emitted by a star in different wavelength regions.

The star Vega is used as the standard reference for the UBV system. By definition, $U = B = V$, which implies $B - V = 0$ and $U - B = 0$. Vega therefore serves as a zero-point reference and is considered approximately white.

The color index provides information about the color and surface temperature of a star.

- If $B - V < 0$, the star emits relatively more blue light than Vega and is therefore **bluer**.
- If $B - V = 0$, the star has a color similar to Vega.
- If $B - V > 0$, the star emits relatively less blue light and appears **redder**.

Since hotter stars emit more radiation at shorter wavelengths, they tend to have smaller or even negative values of $B - V$. Conversely, cooler stars emit more strongly at longer wavelengths and have larger positive values of $B - V$.

The color index is closely related to the surface temperature of a star.

Color Index ($B - V$)	Stellar Type
Negative	Hot, blue stars
Near 0	White stars
Positive	Cool, red stars

Table 1: Relationship between color index and stellar type

Light from distant stars must travel through the interstellar medium before reaching Earth. Interstellar dust absorbs and scatters blue light much more efficiently than red light. As a result, stars observed through dusty regions appear redder than they actually are. This phenomenon is known as **interstellar reddening**.

6.3 Color Excess

The amount of reddening is quantified using the **color excess**. It is defined as the difference between the observed color index and the intrinsic color index of the star, $(B - V)_{\text{obs}} - (B - V)_{\text{int}}$ where

- $(B - V)_{\text{obs}}$ is the observed color index,
- $(B - V)_{\text{int}}$ is the intrinsic color index the star would have in the absence of interstellar absorption.

A positive value of $E(B - V)$ indicates that the star appears redder than its intrinsic color due to extinction by interstellar dust.